

# aiSTROM instructor guide

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Use these materials for teaching or facilitating an AI strategy discussion. The core activity is making and connecting implementation decisions, using the overview and seven dimension sheets. Participants do not need to read the longer references first.

[Public guide](#) · [Blank visual toolkit](#) · [Complete procurement example](#)

## Learning outcomes

Participants should be able to:

- Frame an opportunity before selecting technology.
- Distinguish assumptions, supporting evidence, gaps and unknowns.
- Identify dependencies between data, people, organisation, technology, value, risk and education.
- Turn an unresolved issue into an action with an owner and date.
- Justify a strategic direction: invest, prepare, defer or reject.

## Preparation

Give each group a blank toolkit and a concrete opportunity. Use a real project with shareable information, or one of the fictional exercises in this guide. Print at A3, or confirm that participants can fill, save and reopen the PDFs. Form groups that combine business, domain and technical perspectives where possible.

Ask participants to state their initial proposed approach before showing the completed example. This makes changes in reasoning easier to discuss and reduces copying. Do not ask them to enter confidential organisational information into public files or unapproved AI services.

## A 90-minute session

| Time      | Activity                                                                      | Output                                  |
|-----------|-------------------------------------------------------------------------------|-----------------------------------------|
| 0–10 min  | Frame the problem, beneficiaries and alternatives.                            | Opportunity box and initial proposal.   |
| 10–20 min | Scan the seven dimensions; select the three most consequential uncertainties. | Priority sheets/cards.                  |
| 20–45 min | Work through selected cards. Label assumptions and unknowns.                  | Decisions, strengths, gaps and actions. |
| 45–60 min | Exchange with another group; challenge one unsupported conclusion.            | Revised status or evidence request.     |
| 60–75 min | Connect findings across dimensions and return to the overview.                | Coherent next decision and conditions.  |
| 75–90 min | Debrief using the completed example or group comparisons.                     | Explanation of what changed and why.    |

For 60 minutes, reduce peer review and focus on two or three dimensions. For a longer workshop, give each dimension a separate round and schedule follow-up investigation. Neither format requires resolving all 42 cards in one sitting.

## Facilitation prompts

- What evidence supports this answer, and what are you assuming?
- Would a simpler process or non-AI approach meet the objective?
- If this decision changes, which other sheet must change?
- Is “human review” a named task with time and authority, or just a label?
- What would cause you to revise or stop the proposal?
- What specific action could resolve this unknown, and who owns it?

A card may contain both a strength and a gap. ? Unknown is a useful outcome when evidence is missing. The status controls are not a numerical maturity score. A completed form does not mean the prerequisites are satisfied.

## Discuss the procurement example

The **complete example** is fictional and deliberately unfinished as an investigation: every card is filled, but many statuses remain unknown or gaps.

Trace three connections:

1. **Data 1.2** → **Team 2.1/2.2** → **Technology 4.2**: internal training requires linked cases, expert labels and specialist capacity.
2. **Data 1.4** → **Technology 4.5** → **Risk 6.3**: a service may receive proprietary inputs; provider terms, retention and access affect the build-or-buy choice.
3. **Technology 4.4** → **KPIs 5.3** → **Culture 7.2**: retaining human judgement changes workload; any benefit must account for review work and staff learning.

Ask why the company might strengthen its data and decision process before choosing SaaS, internal development or partnership. Alternative directions are acceptable if their evidence, trade-offs and conditions are coherent. Vendor reviews and trials belong to a later stage.

## Feedback criteria

Assess reasoning rather than the number of boxes filled. Look for a clear problem, appropriate alternatives, traceable evidence, meaningful cross-sheet dependencies, specific actions and a defensible next decision. Invite each group to name one uncertainty it could not resolve.

For professional workshops, agree what records may be retained or shared. A redacted decision summary is often more useful than a complete copy of sensitive project data.

## Workshop exercises

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All scenarios and quantities below are fictional. Their purpose is to expose decisions and dependencies; there is no single required solution.

#### Exercise A: predict equipment failure

A manufacturer wants to predict failures of production equipment. It has two years of sensor readings, repair notes and downtime records. The records use different machine identifiers, several sensors were replaced during that period, and failure labels are incomplete. Maintenance staff currently inspect machines on a schedule.

The proposed system would flag machines for inspection. Missed failures can interrupt production; false alarms consume maintenance time. There is no agreed baseline for either outcome, and newer machines have little historical data.

**Task:** frame the opportunity, compare the existing schedule with a prediction-based approach, and prioritise the Data, Technologies and KPIs sheets. Identify who supplies labels, how to separate training and testing over time, what users need to understand, and how much intervention the team can handle. Connect the findings to Team and Culture.

**Change to introduce halfway through:** the maintenance lead says the team can handle only a few extra inspections per week. Does the proposed metric or threshold still reflect useful performance?

**Debrief:** look for temporal evaluation, capacity-aware value measures, appropriate uncertainty and preservation of operational expertise. An LLM or an agent is not required.

#### Exercise B: should AI support purchasing decisions?

A two-site manufacturer depends on timely replacement parts. Buyers compare supplier quotations against changing engineering specifications. Delays can interrupt production, but the company has not established which causes of delay can be improved by AI. Some records exist; decisions, corrections and delivery outcomes are not consistently linked.

**Task:** start with the broad opportunity. Compare improving current records and rules, buying a service, developing internally and partnering. Use the seven dimension sheets to record upsides, downsides and prerequisites. Decide what strategic direction is justified and what evidence the company should gather before considering offers or trials.

**Change to introduce halfway through:** supplier quotations contain confidential pricing and technical specifications. A proposed service might retain submitted material or use it for model improvement under some account settings. What would the company need to establish before choosing a service? What data and skills would internal training require instead?

**Debrief:** compare strategic routes and dependencies. An agent that contacts suppliers or places orders is only one possible later capability. A completed strategy need not select a technology or vendor.

#### Exercise C: challenge a completed card

Exchange sheets with another group. Select one card marked supported and ask for its evidence. Select one unknown and turn it into a feasible next action. Then identify a decision on another sheet that depends on the answer.

Report one change to the overview. Do not calculate a total score from the status symbols.